

Experiment No. 04

Student Performance Analysis System

Date: 18/09/2026

■ Objective / Aim

To design and implement a **Student Performance Analysis System** in Java using **Object-Oriented Programming (OOP)** principles to calculate marks, total, average, percentage, grade, highest and lowest marks, and overall academic performance.

■■ Problem Statement

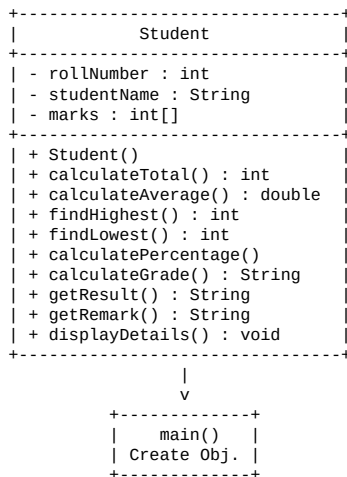
Develop a Java application for a college to analyze the academic performance of a student. The system should accept the student's details and marks and perform the following operations:

- Store subject marks using an array.
- Calculate total and average marks.
- Find highest and lowest marks.
- Calculate percentage.
- Assign a grade based on percentage.
- Determine whether the student has passed or failed.
- Display the student's name using String operations.
- Round percentage to two decimal places using Math.
- Display a performance remark.
- Use a class to represent the student and demonstrate OOP.

■ Concepts Used

Class and Object • Constructor • Arrays • Methods • Loops • Conditional Statements • String Operations • Math Package • Encapsulation • OOP

■ Class Design / UML Diagram



■ Algorithm / Logic Description

1. Create a class named Student.
2. Declare roll number, student name, and marks array.
3. Create a constructor to initialize details.
4. Store five subject marks in an integer array.
5. Use a loop to calculate total marks.
6. Calculate average = Total / Number of Subjects.
7. Use Math.max() and Math.min() to find highest and lowest marks.
8. Calculate percentage from total marks.
9. Use Math.round() to round percentage to two decimal places.
10. Use trim(), toUpperCase(), and length() for the name.
11. Assign grades: 90–100 A+, 80–89 A, 70–79 B, 60–69 C, 50–59 D, below 50 F.
12. Percentage >= 50 means PASS; otherwise FAIL.

13. Display a remark based on grade.
14. Create a Student object in main().
15. Display the complete performance report.

■ Grade Criteria

Percentage	Grade
90 - 100	A+
80 - 89	A
70 - 79	B
60 - 69	C
50 - 59	D
Below 50	F (Fail)

■ Complete Java Source Code

```
class Student {

    private int rollNumber;
    private String studentName;
    private int[] marks;

    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    int calculateTotal() {
        int total = 0;
        for (int mark : marks) {
            total += mark;
        }
        return total;
    }

    double calculateAverage() {
        return (double) calculateTotal() / marks.length;
    }

    int findHighest() {
        int highest = marks[0];
        for (int mark : marks) {
            highest = Math.max(highest, mark);
        }
        return highest;
    }

    int findLowest() {
        int lowest = marks[0];
        for (int mark : marks) {
            lowest = Math.min(lowest, mark);
        }
        return lowest;
    }

    double calculatePercentage() {
        double percentage =
            ((double) calculateTotal() / (marks.length * 100)) * 100;
        return Math.round(percentage * 100.0) / 100.0;
    }

    String calculateGrade() {
        double percentage = calculatePercentage();

        if (percentage >= 90)
            return "A+";
        else if (percentage >= 80)
            return "A";
        else if (percentage >= 70)
            return "B";
        else if (percentage >= 60)
            return "C";
        else if (percentage >= 50)
            return "D";
        else
            return "F";
    }

    String getResult() {
        return calculatePercentage() >= 50 ? "PASS" : "FAIL";
    }

    String getRemark() {
        switch (calculateGrade()) {
            case "A+": return "Outstanding Performance";
            case "A":  return "Excellent Performance";
            case "B":  return "Very Good Performance";
            case "C":  return "Good Performance";
            case "D":  return "Satisfactory Performance";
            default:   return "Needs Improvement";
        }
    }

    void displayDetails() {
```

```

String formattedName = studentName.trim().toUpperCase();

System.out.println("=====");
System.out.println("      STUDENT PERFORMANCE REPORT");
System.out.println("=====");
System.out.println("Roll Number      : " + rollNumber);
System.out.println("Student Name     : " + formattedName);
System.out.println("Name Length      : " + formattedName.length());
System.out.println("-----");
System.out.println("Subject Marks:");

for (int i = 0; i < marks.length; i++) {
    System.out.println("Subject " + (i + 1) + "      : " + marks[i]);
}

System.out.println("-----");
System.out.println("Total Marks      : " + calculateTotal());
System.out.println("Average Marks    : %.2f%n", calculateAverage());
System.out.println("Highest Mark     : " + findHighest());
System.out.println("Lowest Mark      : " + findLowest());
System.out.println("Percentage       : %.2f%%%n", calculatePercentage());
System.out.println("Grade           : " + calculateGrade());
System.out.println("Result          : " + getResult());
System.out.println("Remark          : " + getRemark());
System.out.println("=====");
}
}

public class StudentPerformance {
    public static void main(String[] args) {

        int[] marks = {92, 85, 78, 88, 95};

        Student student = new Student(
            101,
            " rahul kumar ",
            marks
        );

        student.displayDetails();
    }
}

```

■ Compilation and Execution

File name: StudentPerformance.java

```

javac StudentPerformance.java
java StudentPerformance

```

■■ Sample Output

```

=====
      STUDENT PERFORMANCE REPORT
=====
Roll Number      : 101
Student Name     : RAHUL KUMAR
Name Length      : 11
-----
Subject Marks:
Subject 1       : 92
Subject 2       : 85
Subject 3       : 78
Subject 4       : 88
Subject 5       : 95
-----
Total Marks     : 438
Average Marks   : 87.60
Highest Mark    : 95
Lowest Mark     : 78
Percentage      : 87.60%
Grade           : A
Result          : PASS
Remark          : Excellent Performance
=====

```

■ Real-Time Application

College Student Performance Analysis: the system can store subject marks and automatically calculate total, average, percentage, grade, result, and performance remarks.

■ Result

The **Student Performance Analysis System** was successfully implemented in Java using **OOP concepts, arrays, String operations, Math methods, loops, and conditional statements**. The program successfully calculates and displays total marks, average, highest and lowest marks, percentage, grade, result, and performance remark.